

# NAMAN MANOCHA

AI ENGINEER — Applied ML, RAG & Production AI Systems

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## SUMMARY

AI/ML engineer who ships production systems end-to-end — data pipelines, model/retrieval logic, backend APIs, and deployment. Solo-built **FundersAI**, a research platform combining deterministic financial metrics, hybrid retrieval, and explainable ML clustering, and **CareFlow Intelligence**, an agentic RAG + fine-tuning system. Currently building and shipping AI products independently to gain hands-on production experience. Stack: Python/FastAPI, Next.js/React, Supabase/PostgreSQL, Docker/Kubernetes.

## TECHNICAL SKILLS

**ML & AI:** RAG (LangGraph), Hybrid Search, Vector Embeddings (pgvector, ChromaDB), K-Means Clustering, Logistic Regression, MLflow, scikit-learn, Prompt Engineering, Hugging Face, AI Observability (Langfuse)

**Backend & Infra:** Python, FastAPI, REST APIs, SQL, PostgreSQL, Supabase, Redis, Docker, Kubernetes (K3s), GitHub Actions (CI/CD), Google Cloud Run, Vercel

**Frontend & Data:** Next.js, React, TypeScript, ETL Pipelines, Web Scraping, Data Validation & Visualization

## EXPERIENCE

**Technology Manager — PayGain Multiservices Pvt. Ltd.**

Feb 2025 – Present

Kolkata, India

- Scoping and leading development of a company-wide mobile app as sole engineer — client/employee onboarding, document access, product registration, and payments.
- Designed and deployed a lead-capture automation (Make) triggered on Google Form submissions; also drove 30+ client conversions by building and managing the company website and digital discovery flows.

## INDEPENDENT PROJECTS

**FundersAI — AI-Powered Research Platform for Indian Stocks & Mutual Funds**

[fundersai.co.in](https://fundersai.co.in)

Python, FastAPI, Next.js, Supabase/pgvector, LangGraph, Groq, OpenAI, Cloudflare R2 | ~143K LOC, 864 files, 431 commits, 167+ test suites — OpenAI Build Week | Live: [www.fundersai.co.in](https://www.fundersai.co.in) | Repo: [github.com/Naman6019/FundersAI](https://github.com/Naman6019/FundersAI)

- Citation-Grounded RAG:** 7-node LangGraph pipeline (retrieve → grade → synthesize/abstain, with query-rewrite loop) over ~10,400 embedded chunks with lexical reranking and strict LLM abstention — eliminates hallucinated financial claims.
- Explainable ML:** fund-similarity engine using cosine similarity and k-means clustering over 13 standardized features, plus a logistic-regression triage model (MLflow) for parser-review prioritization.
- Data Engineering:** 22-workflow GitHub Actions pipeline scraping/normalizing disclosures from 12 AMCs, reaching 99.66% data-readiness across 871 fund schemes; fixed a normalizer bug affecting 14,000+ records, raising coverage from 83% to 100%.

**CareFlow Intelligence — Agentic RAG & Fine-Tuning System**

Python, FastAPI, Next.js, Supabase/pgvector, ChromaDB, Unsloth, Ollama | ~7,300 LOC, non-clinical, synthetic-data-only

- Agentic RAG + Ingestion:** built a bounded-tool-loop citation-grounded document agent over 4 automated ingestion pipelines (OpenFDA, PubMed, clinical guidelines, media) indexing 37 documents into 620 chunks.
- RAFT Fine-Tuning:** ran a RAFT pipeline (oracle evidence, hard distractors, abstention cases) to fine-tune Qwen2.5-7B via 4-bit QLoRA (Unsloth) — 88.5% citation precision, 92.3% recall — then quantized to GGUF and served locally via Ollama. A second Data Intake Agent enforces human-approval gates on autonomous data ingestion.

## Other Projects

- TalentOS (All Things Agentic Hackathon, Taskmaster Track):** dual-pipeline autonomous agent platform (Google ADK, LangGraph, Firestore) ingesting ~2,500 postings/run across 8 sources with 91% pre-filtering and a 3-state evaluator+drafter chain — ~23K LOC, 131 files, 235+ tests | Live: <https://all-things-agentic--allthingsagentic-505213.asia-southeast1.hosted.app> | Repo: <https://github.com/Naman6019/all-things-agentic>. FundersAI Reports: decoupled LangGraph microservice on a 2-replica Kubernetes Deployment (K3s/AWS EC2). SQuAD QA Benchmarking: fine-tuned/benchmarked BERT/BiDAF/DistilBERT; published as a research paper. GitResume: self-built multi-agent LLM pipeline (PyPI: `git-resume-agent`, ~4K LOC, 25 commits) that synthesizes and adversarially verifies grounded resume bullets from git history.

## EDUCATION & ACHIEVEMENTS

**B.Tech, Computer Science Engineering (AI & ML Specialization) — VIT Amaravati**

June 2021 – Sept 2025

CGPA: 8.2/10 | Published research paper on QA/NLP | Certifications: Azure Administrator Associate, AWS Certified Cloud Practitioner